

ECHS FRAUD ANALYTICS

Advanced Diagnostic Test Overutilisation & Inflation Risk Assessment

CLASSIFICATION	RESTRICTED
ANALYSIS PERIOD	FY 2023-2026 (Data as on 31 Mar 2026)
DATA EXTRACTS ANALYSED	5 CSV extracts + hospital-name and ICD-name mapping files
ANALYTICAL TECHNIQUES	Z-score, robust Z-score, percentile thresholds, provider-density analysis
REPORT PURPOSE	Boardroom / reputed company presentation and audit prioritisation
REPORT DATE	May 2026

PATTERN UNDER INVESTIGATION:

Excessive diagnostic testing, statistical outliers, hospital test-density clusters, ICD justification gaps, repeat beneficiary-hospital loops, and potential rate inflation requiring line-item validation.

EXECUTIVE DASHBOARD - NAMED HOSPITAL AND ICD RISK ASSESSMENT

This revised boardroom report keeps the restricted IIT Kanpur theme and adds hospital names beside hospital IDs and ICD names beside ICD codes. The report removes the earlier weighted score section and focuses on statistically defensible audit triggers: claim-level Z-scores, hospital test-density outliers, repeat referral loops and diagnosis justification gaps.

18,387,688 Diagnostic tests analysed across hospital diagnostic records	18,420,798 Valid claims base excluding CI_HOSPITAL_ID = -1	88 Max tests in one claim 17.6x the median count	915 Repeat referral pairs beneficiary-hospital combinations
12 tests Claim P95 threshold above this requires review	3.35 Hospital P99 density tests per claim threshold	Rs. 19.47 Cr Repeat exposure total amount in repeat pairs	7340 Hospital mapping rows names added where available

Fraud sub-pattern	Key statistical signal	Entities / threshold	Risk decision
Extreme claim-level testing	Z-score up to 8.68; robust Z up to 14.00	8 claims with >= 30 tests	Immediate audit
Hospital diagnostic density	P99 = 3.35 tests/claim; strict band >= 5.0	8 reliable hospitals with >= 100 claims	Priority provider audit
Repeat referral loop	915 pairs; 60.5% same-day repeat gaps	29 pairs with >= 5 referrals	Referral-to-bill matching
Diagnosis justification gap	Top 10 ICD groups = 23.79% of records	ICD names added where available	Clinical appropriateness model
Inflation validation gap	Rate proof requires line-item comparison	hosp_exp_det + inv_proc_rate_list	Next extraction required

Boardroom conclusion: current extracts are strong enough to prioritise overutilisation audit. They are not sufficient to conclude monetary inflation until billed-rate fields are joined with official-rate tables.

1. DATA INTEGRATION, DATABASE SCOPE AND ADDED NAME MAPPING

The revised report uses the original five analytical extracts plus two new mapping inputs: a hospital ID to hospital name workbook and an ICD code to ICD name file. The analysis is intentionally separated into three grains - claim/intimation, hospital/provider, and beneficiary-hospital pair - because each grain detects a different fraud mechanism.

Dataset / table	Key columns used	Why selected	Enhancement in this version
Sheet2 / diag_details extract	DD_INTIMATION_ID, total_test	Direct claim-level diagnostic test count for excessive testing detection	Used for Z-score and robust Z watchlist
Sheet3 / ICD frequency	DD_DIAG_ICD_CODE, frequency	Diagnosis concentration and clinical justification	ICD_Name added using ICD mapping/fallback
Sheet4 hospital aggregate	CI_HOSPITAL_ID, total_claim, total_test, test_per_claim	Provider-level diagnostic density and claim-volume normalisation	Hospital Name, City and Match Status added
Sheet5 repeat referral	CI_BENF_ID, REF_HOSPITAL_ID, REFERRAL_COUNT, total_claim_amt, gap_days	Detects beneficiary-provider loops and financial exposure	Hospital Name added for REF_HOSPITAL_ID where available
Required next join	HED_HOS_RATE, HED_ACT_RATE, HED_CLAIM_AMOUNT, IPRL_NABH_RATE	Proves price/rate inflation through official-rate comparison	Still required for confirmed inflation proof

Hospital mapping coverage

Hospital mapping status	Count	Percentage
MATCHED (DB)	2	0.0%
INFERRED	1,040	14.2%
POLYCLINIC	811	11.0%
PARTIAL (ID)	1,633	22.2%
UNMATCHED	1,176	16.0%
PHONE NUMBER	2,678	36.5%

Important mapping note: some high-risk hospital IDs map only to inferred, partial, unmatched or manual lookup names. The report shows the provided hospital name and match status so that audit users can distinguish confirmed names from master-data reconciliation gaps.

2. STATISTICAL METHODOLOGY - Z-SCORE, ROBUST Z-SCORE AND PERCENTILE RULES

The analytics pipeline uses transparent statistics that can be defended in audit discussions. Weighted score formulas have been removed in this version. The report now uses direct rule-based signals and statistical thresholds instead of a weighted ranking model.

Method	Formula / logic	Why used	Fraud interpretation
Standard Z-score	$Z = (x - \text{mean}) / \text{standard deviation}$	Measures distance from normal population behaviour	$Z > 3$ is a strong outlier; $Z > 5$ is extreme
Robust Z-score	$0.6745 * (x - \text{median}) / \text{MAD}$	More stable under skewed healthcare claims	High robust Z confirms the outlier is not only mean distortion
Percentile thresholds	P95, P99 and maximum	Simple to explain to executives and auditors	Above P99 is rare and audit-worthy
Log-normalised amount review	$\log(1 + \text{amount})$ for amount distribution checks	Reduces the effect of extreme claim amounts	Useful for repeat-referral financial exposure
Rule-based decision bands	Predefined thresholds by claim, hospital and referral behaviour	Avoids black-box scoring	Creates explainable audit queues

Operational principle: a statistical flag is not an allegation of fraud. It is an audit trigger requiring clinical notes, discharge summaries, prescription/test logs and billing-line validation before enforcement.

3. CLAIM-LEVEL DIAGNOSTIC TEST OUTLIER ANALYSIS

Claim-level analysis detects excessive diagnostic testing within a single intimation. The high-risk extract contains 307 claim/intimation records and 2,205 diagnostic tests. The median is 5 tests per claim, while the maximum is 88 tests.

Statistic	Value	Audit meaning
Records analysed	307	High-risk claim/intimation sample
Mean tests per claim	7.18	Average in extracted sample
Median tests per claim	5	Robust normal benchmark
Standard deviation	9.32	Used for Z-score
P95 threshold	12	Review threshold
P99 threshold	56.22	Extreme statistical threshold
Maximum	88	Top claim = 17.6x median

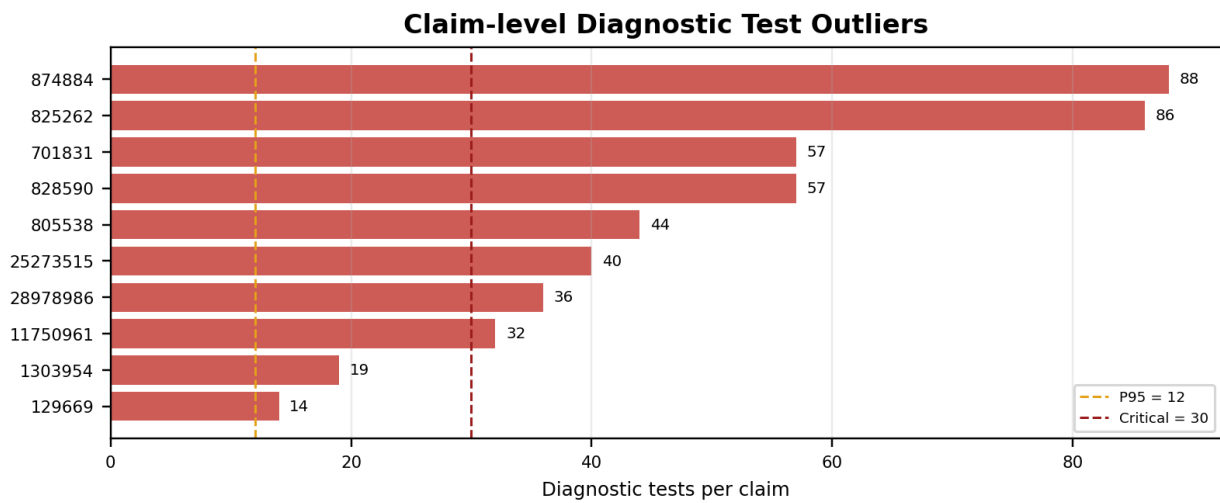


Figure 3.1 - Top DD_INTIMATION_ID records ranked by diagnostic test count with P95 and critical thresholds.

3A. CLAIM-LEVEL WATCHLIST WITH Z-SCORE VALIDATION						
Rank	DD_INTIMATION_ID	Tests	Z-score	Robust Z	Risk	Recommended action
1	874884	88	8.67	14.00	CRITICAL	Immediate audit
2	825262	86	8.45	13.66	CRITICAL	Immediate audit
3	701831	57	5.34	8.77	CRITICAL	Immediate audit
4	828590	57	5.34	8.77	CRITICAL	Immediate audit
5	805538	44	3.95	6.58	CRITICAL	Immediate audit
6	25273515	40	3.52	5.90	CRITICAL	Immediate audit
7	28978986	36	3.09	5.23	CRITICAL	Immediate audit
8	11750961	32	2.66	4.55	CRITICAL	Immediate audit
9	1303954	19	1.27	2.36	HIGH	Clinical validation
10	129669	14	0.73	1.52	HIGH	Clinical validation
11	10129828	13	0.62	1.35	HIGH	Clinical validation
12	10858772	13	0.62	1.35	HIGH	Clinical validation

Interpretation for senior meeting

The top two claims have Z-scores above 8 and robust Z-scores above 13. This means the records remain extreme even after using median-based robust statistics that reduce the influence of skewed healthcare distributions.

Audit action: all claims with total_test >= 30 should be cross-checked against admission duration, diagnosis, discharge summary, repeated test dates, package rules and line-item billing.

Claim-level operational rules

Rule	Condition	Risk tier	Action
Excessive testing	total_test > 12	HIGH	Audit review
Critical testing	total_test >= 30	CRITICAL	Immediate specialist review
Statistical outlier	z_test_count > 3	CRITICAL	Payment hold / sample validation
Extreme robust outlier	robust_z_test_count > 5	CRITICAL	Investigate duplicate tests and panels

4. HOSPITAL-LEVEL DIAGNOSTIC DENSITY ANALYSIS

Hospital-level overutilisation is best measured through diagnostic tests per claim, not raw test volume. Raw volume favours large hospitals, while test density normalises for provider size. This version adds hospital names beside hospital IDs wherever available in the uploaded mapping workbook.

Statistic	Value	Fraud relevance
Hospitals with diagnostic records	7,341	Provider universe for density scoring
Total diagnostic tests	18,387,688	Total diagnostic activity
Mean tests per claim	1.26	Average provider density
Median tests per claim	1.00	Robust baseline
P95 density	2.00	Monitoring threshold
P99 density	3.35	High-risk audit threshold
Maximum density	8.00	Maximum observed test-per-claim

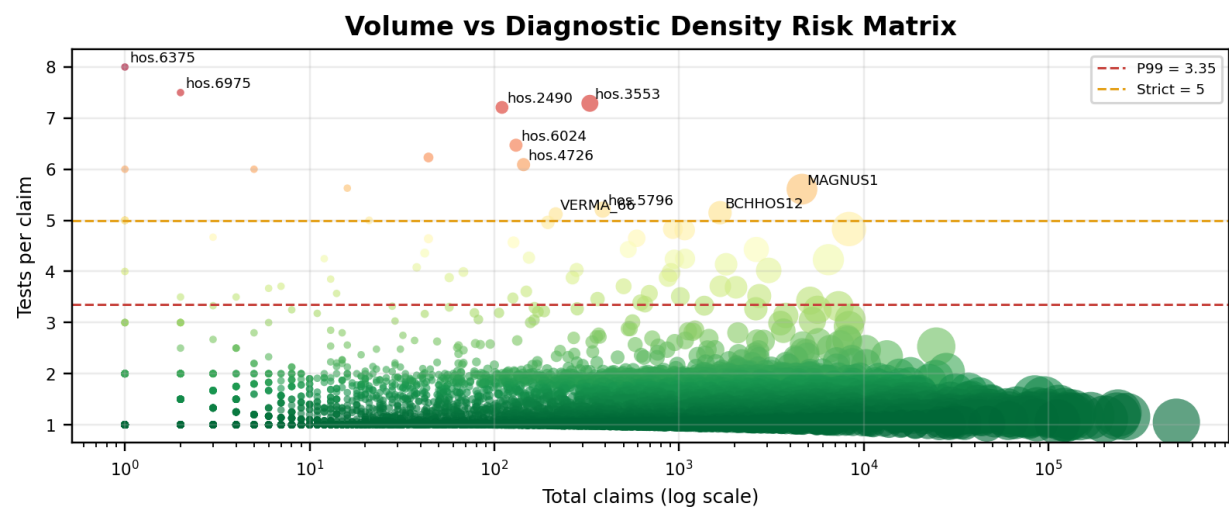


Figure 4.1 - Volume vs diagnostic density matrix. High y-axis plus sufficient claim volume is the priority audit zone.

4A. RELIABLE HIGH-RISK HOSPITAL DENSITY OUTLIERS WITH HOSPITAL NAMES

Small-denominator hospitals can show unstable ratios. Therefore, the strongest provider-level cases are filtered to hospitals with meaningful claim volume. The strict decision rule is test_per_claim > 5 and total_claims >= 100.

Rank	Hospital ID	Hospital Name	Match Status	Claims	Tests	Tests/claim	Density Z	Risk
1	hos.3553	Hospital ID #3553 - lookup ...	PARTIAL (ID)	329	2,400	7.29	11.80	CRITICAL
2	hos.2490	Hospital ID #2490 - lookup ...	PARTIAL (ID)	110	793	7.21	11.65	CRITICAL
3	hos.6024	Hospital ID #6024 - lookup ...	PARTIAL (ID)	131	847	6.47	10.20	CRITICAL
4	hos.4726	Hospital ID #4726 - lookup ...	PARTIAL (ID)	144	877	6.09	9.45	CRITICAL
5	MAGNUS1	Unknown - Manual Lookup Req...	UNMATCHED	4,630	25,962	5.61	8.51	CRITICAL
6	hos.5796	Hospital ID #5796 - lookup ...	PARTIAL (ID)	386	2,011	5.21	7.73	CRITICAL
7	BCHHOS12	BCH HOSPITAL	INFERRED	1,667	8,593	5.15	7.61	CRITICAL
8	VERMA_66	VERMA HOSPITAL	INFERRED	215	1,100	5.12	7.55	CRITICAL

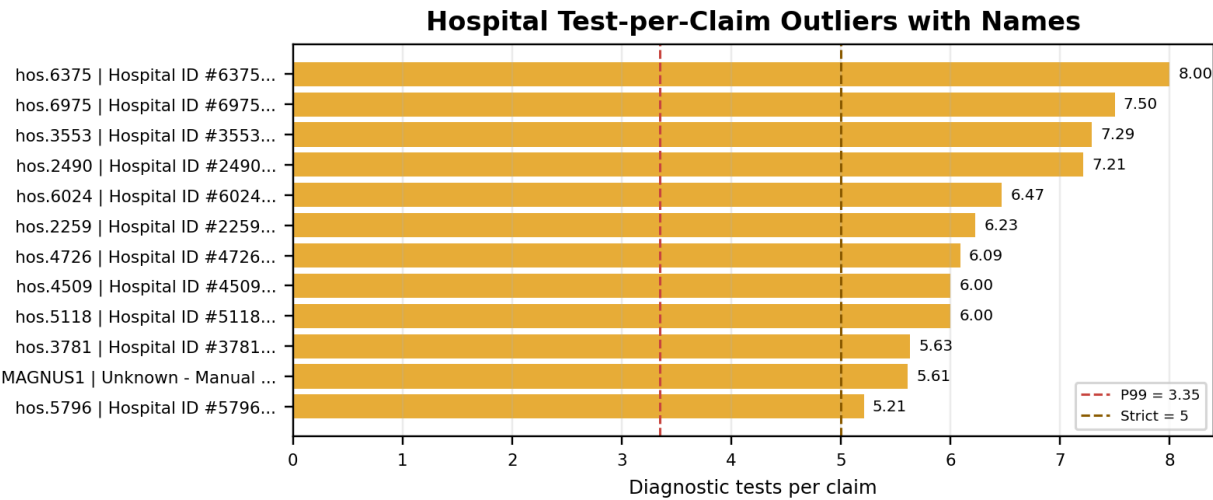


Figure 4.2 - Highest test-per-claim hospitals with hospital names shown in chart labels.

Provider conclusion: high-density hospitals with sufficient claim volume should be checked for test bundling, duplicate tests, unnecessary panels, diagnosis-test mismatch and package rule deviations.

5. REPEAT BENEFICIARY-HOSPITAL REFERRAL LOOP ANALYSIS

Repeat referral behaviour detects a different fraud mechanism: the same beneficiary repeatedly receiving referrals to the same hospital, potentially generating repeated diagnostic claims. This is especially important where the gap is zero days or where total exposure exceeds Rs. 10 lakh.

Statistic	Value	Fraud relevance
Repeat beneficiary-hospital pairs	915	Loop universe
Total financial exposure	Rs. 194,700,696	Aggregate repeat referral amount
Mean amount per pair	Rs. 212,788	Average exposure
Median amount per pair	Rs. 73,815	Robust central value
P95 amount	Rs. 890,282	High-value threshold
P99 amount	Rs. 1,481,413	Extreme value threshold
Maximum pair amount	Rs. 3,315,447	Highest repeat-pair exposure
Pairs with 5+ referrals	29	High-frequency loop flag
Same-day repeat gaps	554 (60.5%)	Duplicate / fragmentation red flag

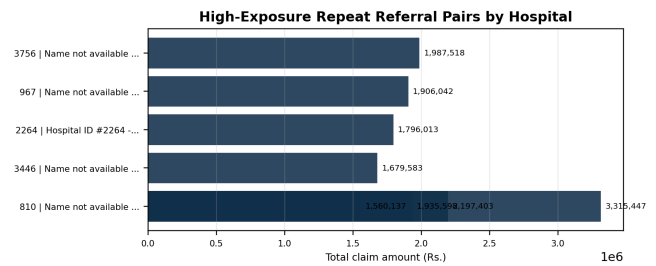
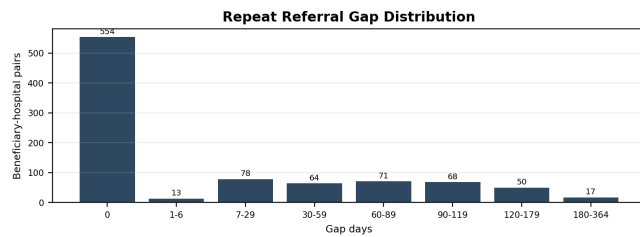


Figure 5.1 - Same-day referral gaps and high-exposure repeat referral pairs with hospital names when mapped.

5A. HIGH-EXPOSURE REPEAT REFERRAL WATCHLIST WITH HOSPITAL NAMES								
Rank	Beneficiary ID	Hospital ID	Hospital Name	Refs	Total claim amount	Avg claim	Gap days	Risk
1	4334007	810	Name not available in provi...	4	Rs. 3,315,447	Rs. 828,862	104	CRITICAL
2	1983604	810	Name not available in provi...	5	Rs. 2,197,403	Rs. 439,481	111	CRITICAL
3	3847538	3756	Name not available in provi...	2	Rs. 1,987,518	Rs. 993,759	0	CRITICAL
4	2400512	810	Name not available in provi...	4	Rs. 1,935,598	Rs. 483,900	121	CRITICAL
5	3958561	967	Name not available in provi...	4	Rs. 1,906,042	Rs. 476,510	165	CRITICAL
6	2318645	2264	Hospital ID #2264 - lookup ...	5	Rs. 1,796,013	Rs. 359,203	125	CRITICAL
7	4025576	3446	Name not available in provi...	3	Rs. 1,679,583	Rs. 559,861	96	CRITICAL
8	4351918	810	Name not available in provi...	3	Rs. 1,560,137	Rs. 520,046	92	CRITICAL
9	3769477	3800	Hospital ID #3800 - lookup ...	3	Rs. 1,537,656	Rs. 512,552	0	CRITICAL
10	1885966	452	Name not available in provi...	2	Rs. 1,483,444	Rs. 741,722	0	CRITICAL
11	4311578	3677	Hospital ID #3677 - lookup ...	3	Rs. 1,468,938	Rs. 489,646	44	CRITICAL
12	4143044	3579	Hospital ID #3579 - lookup ...	3	Rs. 1,455,349	Rs. 485,116	96	CRITICAL

Risk amplifier

Hospital ID 810 appears repeatedly in the highest-exposure repeat referral cases. Its hospital name is not available in the provided mapping file, so it remains a master-data reconciliation priority before operational deployment.

Audit action: high-value repeat pairs should be reviewed through referral-to-bill matching, diagnostic procedure review, beneficiary history review and doctor/referral-source mapping.

Repeat referral rules			
Rule	Condition	Risk tier	Reason
Repeat loop	referral_count >= 5	HIGH	Repeated beneficiary-hospital cycle
High-value repeat pair	total_claim_amt >= Rs. 10 lakh	CRITICAL	Material exposure
Same-day repeat	gap_days = 0	HIGH	Possible duplicate or fragmented authorisation
Extreme frequency	referral_count >= 10	CRITICAL	Unusual repeat behaviour

6. ICD / DIAGNOSIS JUSTIFICATION ANALYSIS WITH ICD NAMES

Diagnosis information is necessary because diagnostic intensity must be judged against clinical context. A high test count may be valid for oncology, nephrology or critical care, but suspicious for routine OPD or low-complexity diagnoses. This version adds ICD_Name beside each ICD code.

Statistic	Value	Interpretation
ICD codes / groups	3,079	Clinical grouping universe
Total diagnosis records	1,136,515	Diagnosis volume
Mean frequency	369.12	Average diagnosis count
Median frequency	35	Robust baseline
P99 frequency	7,139.14	Extreme diagnosis concentration
Top 10 share	23.79%	Concentration in leading codes/groups
Top 20 share	36.95%	Broad diagnosis bands require standardisation

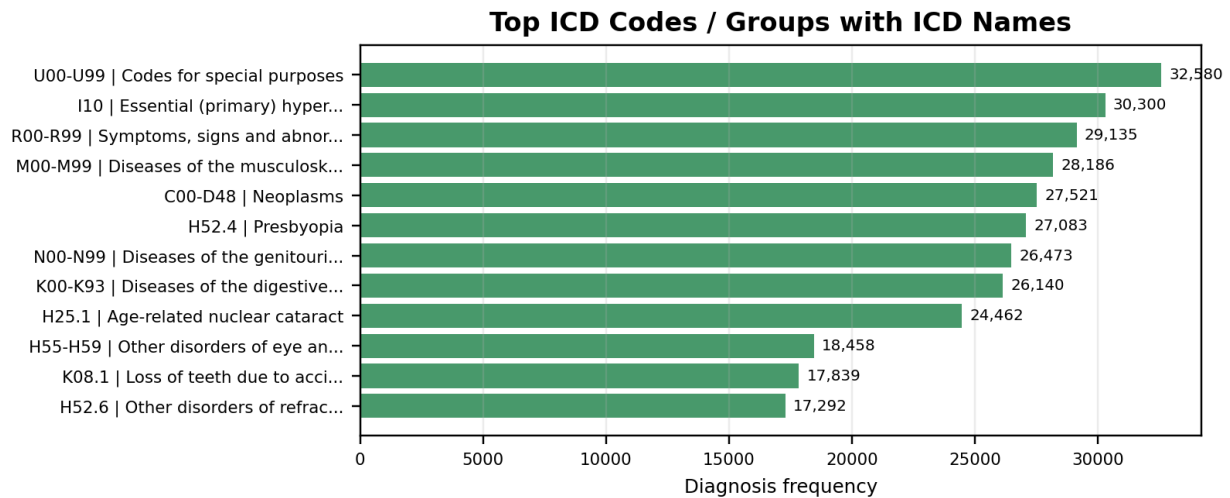


Figure 6.1 - Top ICD / diagnosis groups with ICD names. Broad ICD ranges need mapping before model deployment.

6A. DIAGNOSIS NORMALISATION AND ICD NAME MAPPING

Rank	ICD Code / Group	ICD_Name	Frequency	Share	Z-frequency	Type	Name source
1	U00-U99	Codes for special purposes	32,580	2.87%	16.54	Broad group	Standard ICD-10 fallback
2	I10	Essential (primary) hypertension	30,300	2.67%	15.36	Specific code	Standard ICD-10 fallback
3	R00-R99	Symptoms, signs and abnormal clinical/l...	29,135	2.56%	14.77	Broad group	Standard ICD-10 fallback
4	M00-M99	Diseases of the musculoskeletal system ...	28,186	2.48%	14.28	Broad group	Standard ICD-10 fallback
5	C00-D48	Neoplasms	27,521	2.42%	13.94	Broad group	Standard ICD-10 fallback
6	H52.4	Presbyopia	27,083	2.38%	13.71	Specific code	Standard ICD-10 fallback
7	N00-N99	Diseases of the genitourinary system	26,473	2.33%	13.40	Broad group	Standard ICD-10 fallback
8	K00-K93	Diseases of the digestive system	26,140	2.30%	13.23	Broad group	Standard ICD-10 fallback
9	H25.1	Age-related nuclear cataract	24,462	2.15%	12.37	Specific code	Standard ICD-10 fallback
10	H55-H59	Other disorders of eye and adnexa	18,458	1.62%	9.29	Broad group	Standard ICD-10 fallback
11	K08.1	Loss of teeth due to accident, extracti...	17,839	1.57%	8.97	Specific code	Standard ICD-10 fallback
12	H52.6	Other disorders of refraction	17,292	1.52%	8.69	Specific code	Standard ICD-10 fallback

Recommended diagnosis-test outlier feature

$$\text{diagnosis_test_outlier} = \text{actual_test_count} - \text{expected_test_count_for_same_ICD_group}$$

The expected test count should be calculated within peer groups such as ICD category, age band, admission type, hospital type/accreditation and claim route. This avoids unfairly flagging clinically complex cases while still detecting abnormal testing for routine diagnoses.

Data quality note: the uploaded ICD master contains 300 rows and is concentrated in A00-A54.5. Several top ICD groups in the diagnostic extract are not present in that file; for presentation, standard ICD-10 descriptions were used as a fallback and the source is labelled in the table.

7. RECOMMENDED FRAUD DETECTION RULE SET

This version avoids weighted scoring and uses a direct rule engine. The rules are defensible because they are based on Z-scores, robust statistics, percentile thresholds, high-value exposure and repeat referral behaviour.

Rule family	Condition	Risk tier	Reason / action
Claim excessive testing	total_test > 12	HIGH	Above P95; audit review
Claim critical testing	total_test >= 30	CRITICAL	Immediate specialist review
Claim Z-score outlier	z_test_count > 3	CRITICAL	Strong statistical outlier
Hospital density outlier	test_per_claim > 3.35	HIGH	Above hospital P99
Reliable provider outlier	test_per_claim > 5 AND total_claims >= 100	CRITICAL	High density with sufficient volume
Repeat referral loop	referral_count >= 5	HIGH	Repeated beneficiary-hospital cycle
High-value repeat pair	total_claim_amt >= Rs. 10 lakh	CRITICAL	Material financial exposure
Same-day repeat	gap_days = 0	HIGH	Duplicate / fragmented referral red flag
Diagnosis mismatch	actual tests > expected tests by ICD peer group	HIGH	Clinical justification gap
Rate inflation	rate_inflation_ratio > policy threshold	CRITICAL	Official-rate violation

Positioning for reputed-company presentation: the current model should be described as overutilisation screening and audit prioritisation. Monetary inflation must be confirmed only after line-item billing and official-rate data are joined.

8. FOLLOW-UP DATA EXTRACTION QUERIES FOR VALIDATION

The next extraction should move from aggregate analysis to line-item validation. The following SQL patterns are recommended for database extraction and audit-ready evidence generation.

```

Claim-level excessive diagnostic testing
SELECT hed.HED_INTIMATION_ID, COUNT(*) AS diagnostic_line_count,
       SUM(COALESCE(hed.HED_CLAIM_UNITS,1)) AS diagnostic_units,
       SUM(hed.HED_CLAIM_AMOUNT) AS claimed_amt,
       SUM(hed.HED_APP_AMOUNT) AS approved_amt
FROM hosp_exp_det hed
WHERE hed.HED_PROC_ID IS NOT NULL
GROUP BY hed.HED_INTIMATION_ID
HAVING diagnostic_line_count > 12 OR diagnostic_units > 12
ORDER BY diagnostic_units DESC;

Hospital diagnostic density with names after master reconciliation
SELECT ci.CI_HOSPITAL_ID, om.OM_OFFICE_NAME AS hospital_name,
       COUNT(DISTINCT ci.CI_INTIMATION_ID) AS total_claims,
       COUNT(*) AS diagnostic_lines,
       ROUND(COUNT(*) / COUNT(DISTINCT ci.CI_INTIMATION_ID),2) AS tests_per_claim
FROM claim_intimation ci
JOIN hosp_exp_det hed ON ci.CI_INTIMATION_ID = hed.HED_INTIMATION_ID
LEFT JOIN office_master om ON om.OM_OFFICE_ID = ci.CI_HOSPITAL_ID
WHERE ci.CI_HOSPITAL_ID IS NOT NULL AND ci.CI_HOSPITAL_ID <> '-1'
GROUP BY ci.CI_HOSPITAL_ID, om.OM_OFFICE_NAME
HAVING total_claims >= 100 AND tests_per_claim > 3.35
ORDER BY tests_per_claim DESC;

ICD name enrichment
SELECT dd.DD_DIAG_ICD_CODE, im.ICD_DESC AS ICD_Name, COUNT(*) AS frequency
FROM diag_details dd
LEFT JOIN icd_master im ON im.ICD_CODE = dd.DD_DIAG_ICD_CODE
GROUP BY dd.DD_DIAG_ICD_CODE, im.ICD_DESC
ORDER BY frequency DESC;

Rate inflation validation
SELECT hed.HED_INTIMATION_ID, ci.CI_HOSPITAL_ID, hed.HED_PROC_ID,
       hed.HED_PROC_DESC, hed.HED_HOS_RATE, hed.HED_ACT_RATE, hed.HED_CLAIM_AMOUNT,
       r.IPRL_NABH_RATE, r.IPRL_NON_NABH_RATE,
       ROUND(hed.HED_HOS_RATE / NULLIF(r.IPRL_NABH_RATE,0),2) AS rate_inflation_ratio
FROM hosp_exp_det hed
JOIN claim_intimation ci ON ci.CI_INTIMATION_ID = hed.HED_INTIMATION_ID
LEFT JOIN inv_proc_rate_list r ON r.IPRL_ID = hed.HED_PROC_ID
WHERE hed.HED_HOS_RATE > 0
ORDER BY rate_inflation_ratio DESC;

```

9. CONCLUSIONS, GOVERNANCE AND PRESENTATION TALK TRACK

Conclusion	Evidence	Business implication
C1. Claim-level outliers are statistically extreme	Max 88 tests; Z-score 8.68; robust Z 14.00	Immediate audit of top 8 intimation IDs
C2. Provider density outliers are reliable	8 hospitals with test_per_claim > 5 and >= 100 claims	Provider audit, test-panel validation, package rule checks
C3. Repeat referral loops are material	915 pairs; Rs. 19.47 Cr exposure; 60.5% same-day gaps	Referral-to-bill matching and beneficiary-provider loop review
C4. ICD names improve clinical interpretability	ICD_Name added beside top ICD codes/groups	Expected-test-by-diagnosis model becomes explainable
C5. Monetary inflation requires line-item proof	hosp_exp_det and inv_proc_rate_list needed	Do not allege price inflation without official-rate comparison

Recommended meeting narrative

This is an advanced statistical fraud-risk assessment. We have identified clear overutilisation signals using transparent and defensible statistics. The revised report improves interpretability by adding hospital names and ICD names while removing weighted score pages. The next step is targeted audit sampling, clinical validation and rate-line extraction for inflation proof.

Governance safeguards

Safeguard	Purpose
Human-in-the-loop audit	Avoid automatic enforcement from statistical flags
Clinical document validation	Confirm medical necessity before disallowance
Hospital master reconciliation	Resolve numeric and alphanumeric hospital IDs and missing names
ICD master completion	Ensure all ICD codes/groups have standard clinical descriptions
Rate-table validation	Separate overuse from overpricing
Feedback-labelled model improvement	Use audit outcomes to train future predictive models

Final executive statement: the current datasets prove high-value overutilisation risk and support immediate targeted audit. Confirmed monetary inflation requires the next-stage join to billing line items and official rate schedules.

- End of Report -